TEACHER EDITION

Motion and force. A 60-minute lesson.

CBSE Class 9 Physics. Each page shows one slide beside what to say, what to ask, and the misconception to watch for.

BEFORE THE LESSON

Have a book, a coin, a playing card and a glass ready for quick demonstrations. Write the three SI units, m, kg and s, in a corner of the board.

Syllabus: aligned to CBSE Class 9 Science 2026–27, NCERT *Exploration* chapters “Describing Motion Around Us” and “How Forces Affect Motion”. Physical quantities, SI units and scalars and vectors are taught as foundations.

00	Opening	0–3 min
01	What is physics?	3–7 min
02	Physical quantities and SI units	7–13 min
03	Scalars and vectors	13–17 min
04	Distance and displacement	17–23 min
05	Speed and velocity	23–30 min
06	Uniform and non-uniform motion	30–35 min
07	Acceleration	35–40 min
08	Equations of motion	40–48 min
09	Force and Newton's laws	48–57 min
10	Practice and recap	57–60 min
	Total	**60 min**

Practice questions 3 to 8 and the answer key are set as homework.

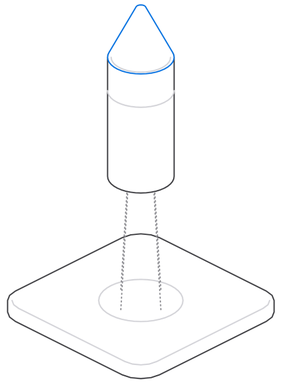
Minute 0 · 1 min

CBSE CLASS 9 · PHYSICS · 60-MINUTE LESSON

Motion and force.

Physics from the very first step: how things move, how we measure it, and why it happens.

UNITS · VECTORS · SPEED AND VELOCITY · ACCELERATION · NEWTON'S LAWS



CBSE Class 9 · Physics · Motion and force

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SAY

Welcome the class. Today is about the most visible part of physics: things moving. By the end, everyone will be able to describe any motion with numbers and explain what causes it.

ASK

Before we start: what is the fastest thing you have seen move today?

Minute 1 · 1 min

00 Opening

Minute 1

THE PLAN

Sixty minutes, ten small steps.

01	What is physics?	3–7 min	02	Physical quantities and SI units	7–13 min
03	Scalars and vectors	13–17 min	04	Distance and displacement	17–23 min
05	Speed and velocity	23–30 min	06	Uniform and non-uniform motion	30–35 min
07	Acceleration	35–40 min	08	Equations of motion	40–48 min
09	Force and Newton's laws	48–57 min	10	Practice and recap	57–60 min

Aligned to CBSE Class 9 Science 2026–27 (NCERT *Exploration*: “Describing Motion Around Us” and “How Forces Affect Motion”). Units and vectors are taught as foundations.

CBSE Class 9 · Physics · Motion and force

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SAY

Walk through the map in under a minute. Point out the pattern: we start with words, move to measuring, then to calculating, and finish with what causes motion.

Minute 2 · 1 min

00 Opening

Minute 2

HOW WE WILL LEARN

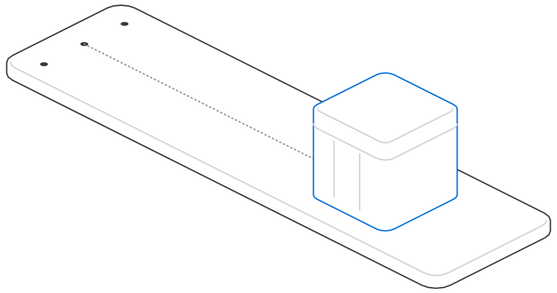
Every idea gets the same four steps.

01 **A simple definition**
In words you could explain to a friend.

02 **An everyday example**
Cars, bicycles, balls and you.

03 **A picture and a formula**
With every symbol and unit explained.

04 **A worked problem**
Step by step, then you try one.



SAY

Set expectations: no one needs to be good at maths already. Every formula will be built from an everyday idea first.

MISCONCEPTION TO ADDRESS

Some students believe physics is 'just formulas'. Stress that each formula is a sentence about the world written in short form.

Minutes 3–7 · 2 min

01 What is physics?

Minutes 3–7

01 · WHAT IS PHYSICS?

Physics is the study of matter, energy, and how they interact.

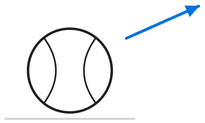
It explains how things move and why. The same few rules work for a bicycle, a cricket ball and a rocket.



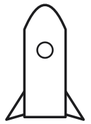
Cycling to school



Braking a car



Bowling a ball



Launching a rocket

SAY

Give the definition in one breath, then make it concrete: physics answers two questions about anything that moves. How is it moving? Why is it moving that way? Point at each picture and ask which question it raises.

ASK

Which of these four would be hardest to stop, and why?

MISCONCEPTION TO ADDRESS

Students often think physics is only about laboratories. Every picture here is something they do or see daily.

Minutes 3–7 · 2 min

01 What is physics?

Minutes 3–7

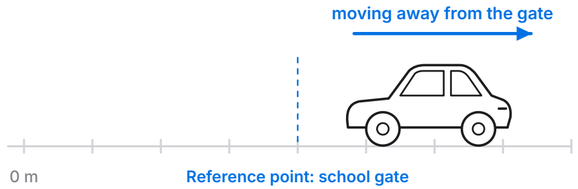
MOTION IS EVERYWHERE

Describing motion needs three things.

A reference point
Where are we measuring from? A school gate, a lamp post, the start line.

A measurement
How far, how fast, how long, with a unit.

A direction
Which way? East, up, towards the wall.



ASK THE CLASS
Is a student sitting at a desk moving? Think about the Earth.

SAY

Motion only makes sense compared with something: a reference point. Use the car: it is moving relative to the school gate. Then ask the question on the slide.

ASK

Is a student sitting at a desk moving? (Answer: relative to the classroom, no. Relative to the Sun, yes, because the Earth moves. Motion is relative.)

MISCONCEPTION TO ADDRESS

Many students think motion is absolute. Address it here: whether something is moving depends on the reference point.

Minutes 7–13 · 1.5 min

02 Physical quantities and SI units

Minutes 7–13

02 · PHYSICAL QUANTITIES

A measurement is a number and a unit.

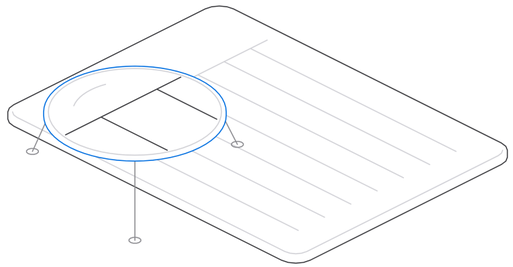
DEFINITION

A **physical quantity** is anything we can measure, such as length, mass or time.

5
means nothing

→

5 m
five metres



SAY

Hold up a pencil. Ask 'how long is it?' If a student says '15', ask '15 what?' That is the whole point: a number without a unit is not a measurement.

ASK

If I say my school is '2' from home, what could I mean? (2 km? 2 minutes? 2 bus stops?)

MISCONCEPTION TO ADDRESS

Students drop units in answers. From today, every answer needs a unit, and marks depend on it in exams.

Minutes 7–13 · 1.5 min

THE SI SYSTEM

Seven base units. Everything else is built from them.

SI stands for the International System of Units, used by scientists in every country.

QUANTITY	UNIT	SYMBOL
Length	metre	**m**
Mass	kilogram	**kg**
Time	second	**s**
Electric current	ampere	**A**
Temperature	kelvin	**K**
Amount of substance	mole	**mol**
Luminous intensity	candela	**cd**

TODAY WE NEED THREE

Length in **metres**, mass in **kilograms** and time in **seconds**. Speed, acceleration and force are all built from these.

Watch out: the symbol for second is **s**, not "sec". Units named after people, like ampere, still use a small letter when written in full.

SAY

Do not ask students to memorise all seven today. Highlight the first three. Mention that the kilogram is the only base unit with a prefix built in.

MISCONCEPTION TO ADDRESS

Common errors: writing 'Kg' or 'Sec', and mixing units such as km with seconds in one formula.

Minutes 7–13 · 2 min

02 Physical quantities and SI units

Minutes 7–13

DERIVED UNITS AND PREFIXES

Bigger and smaller units.

PREFIX	SYMBOL	MEANS
kilo	k	× 1000
centi	c	÷ 100
milli	m	÷ 1000

Speed is a **derived** quantity: distance ÷ time. Its SI unit is the **metre per second**, $\mathrm{m/s}$ or $\mathrm{m\ s^{-1}}$.

THE CONVERSION YOU WILL USE MOST

$$1\ \mathrm{km/h} = \frac{1000\ \mathrm{m}}{3600\ \mathrm{s}} = \frac{5}{18}\ \mathrm{m/s}$$

km/h to m/s: multiply by $\frac{5}{18}$. m/s to km/h: multiply by $\frac{18}{5}$.

WORKED EXAMPLE

QUESTION — A car moves at 72 km/h. What is this in m/s?

FORMULA — $v\ (\mathrm{m/s}) = v\ (\mathrm{km/h}) \times \frac{5}{18}$

SOLVE — $72 \times \frac{5}{18} = 4 \times 5$

ANSWER — $20\ \mathrm{m/s}$

SAY

Build the 5/18 factor on the board: 1 km is 1000 m, 1 hour is 3600 s, so 1000/3600 simplifies to 5/18. Students remember a factor they derived far better than one they were told.

ASK

Quick: what is 36 km/h in m/s? (10 m/s.)

MISCONCEPTION TO ADDRESS

Students multiply by 18/5 when they should multiply by 5/18. Tip: m/s numbers are always smaller than the same speed in km/h.

Minutes 7–13 · 1 min

02 Physical quantities and SI units

Minutes 7–13

YOUR TURN

Two quick conversions.

QUESTION 1

A cyclist rides at $18\ \mathrm{km/h}$. What is that in m/s?

QUESTION 2

A cheetah can run at about $30\ \mathrm{m/s}$. What is that in km/h?

SAY

Give 60 seconds. Take answers from two students and ask them to say the factor they used.

ASK

Answers: 18 × 5/18 = 5 m/s. 30 × 18/5 = 108 km/h.

Minutes 13–17 · 2 min

03 Scalars and vectors

Minutes 13–17

03 · SCALARS AND VECTORS

Some quantities need a direction.

SCALAR

Has **size only**. Mass, time, distance, speed, temperature.

VECTOR

Has **size and direction**. Displacement, velocity, acceleration, force.

We draw a vector as an arrow: its length shows the size, its head shows the direction.



"5 km" is a scalar. "5 km north" is a vector.

SAY

Use directions to a friend's house: '5 km' is not enough to find it, '5 km north' is. That is the difference between a scalar and a vector.

MISCONCEPTION TO ADDRESS

Students think vectors are only arrows on paper. Stress that the direction is part of the quantity itself.

Minutes 13–17 · 2 min

03 Scalars and vectors

Minutes 13–17

SORT THEM

Scalar or vector?

Mass	Velocity	Time	Force
of a school bag	of a train going west	taken for a race	pushing a door open

ASK THE CLASS

Hold up one finger for scalar, two for vector, as I point to each card.

SAY

Point to each card in turn and let the class vote with fingers. Then reveal: mass scalar, velocity vector, time scalar, force vector.

ASK

Answers: Mass: scalar. Velocity: vector. Time: scalar. Force: vector (a push has a direction).

MISCONCEPTION TO ADDRESS

Some students say time is a vector because it 'goes forwards'. In Class 9 physics, time is a scalar.

Minutes 17–23 · 2 min

04 Distance and displacement

Minutes 17–23

04 · DISTANCE AND DISPLACEMENT

How far you went, and how far you got.

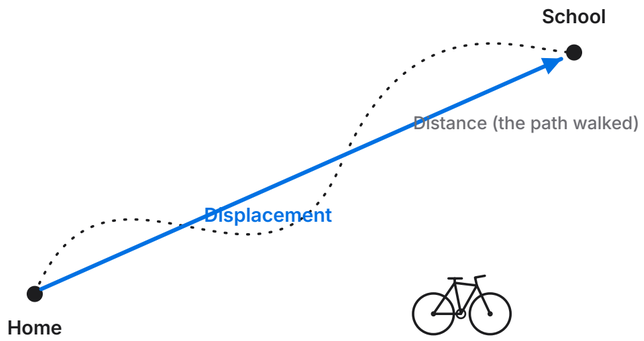
DISTANCE

The **total length of the path** travelled. A scalar.

DISPLACEMENT

The **shortest straight line** from start to finish, with direction. A vector.

Both are measured in metres (m).



SAY

Ask a student how they come to school. The road twists: that is distance. Now imagine a bird flying straight from home to school: that straight arrow is displacement.

ASK

If you walk to the canteen and back to class, what is your displacement? (Zero: you ended where you started.)

MISCONCEPTION TO ADDRESS

Students think displacement is just 'distance with a direction'. It is the straight-line change in position, so it can be much smaller than distance, even zero.

04 Distance and displacement

Minutes 17–23

WORKED EXAMPLE

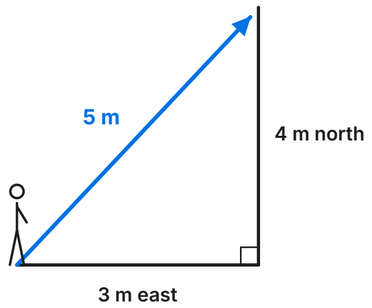
A walk around the playground.

QUESTION — Riya walks 3 m east, then 4 m north. Find the distance and the displacement.

DISTANCE — Add the path lengths: $3 + 4 = 7\ \text{m}$

DISPLACEMENT — The straight line is the hypotenuse: $\sqrt{3^2 + 4^2} = \sqrt{25}$

ANSWER — **Distance $= 7\ \text{m}$. Displacement $= 5\ \text{m}$ towards the north-east.**



SAY

Draw the right angle on the board. Most Class 9 students know Pythagoras from maths: link it explicitly. Emphasise that a displacement answer needs a direction.

ASK

Which is bigger, and can it ever be the other way round?

MISCONCEPTION TO ADDRESS

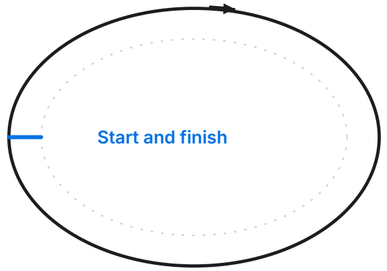
Students add 3 + 4 for displacement. Ask: 'Did Riya end up 7 m from where she started?'

Minutes 17–23 · 2 min

WORKED EXAMPLE

One lap of the track.

QUESTION	An athlete runs exactly one lap of a 400 m track and stops at the start line.
DISTANCE	The whole lap was run: 400 m.
DISPLACEMENT	Start and finish are the same point, so the straight line between them has zero length.
ANSWER	**Distance** $= 400\ \mathrm{m}$. **Displacement** $= 0\ \mathrm{m}$.



ASK THE CLASS

Can displacement ever be greater than distance?

SAY

This is the example students remember. Run your finger round the track, then point at the start: 'How far is she from where she began?'

ASK

Can displacement ever be greater than distance? (No. It can be equal, when the path is a straight line in one direction, or smaller, but never greater.)

MISCONCEPTION TO ADDRESS

Students think zero displacement means 'did not move'. She ran 400 m; she simply came back.

Minutes 23–30 · 1.5 min

05 Speed and velocity

Minutes 23–30

05 · SPEED

How fast something covers distance.

DEFINITION

Speed is the distance travelled per unit time. A scalar.

$$\text{speed} = \frac{\text{distance}}{\text{time}} \qquad v = \frac{s}{t}$$

SYMBOL	MEANS	SI UNIT
v	speed	m/s
s	distance travelled	m
t	time taken	s



km/h

A speedometer shows the speed at each instant.

SAY

Speed is something students already feel. Link to the speedometer: it does not care which way the car points, only how fast it goes. That is why speed is a scalar.

MISCONCEPTION TO ADDRESS

Note that in the speed formula, s means distance here; later, in velocity and the equations of motion, s means displacement. Say this out loud to avoid confusion.

Minutes 23–30 · 1.5 min

05 Speed and velocity

Minutes 23–30

VELOCITY

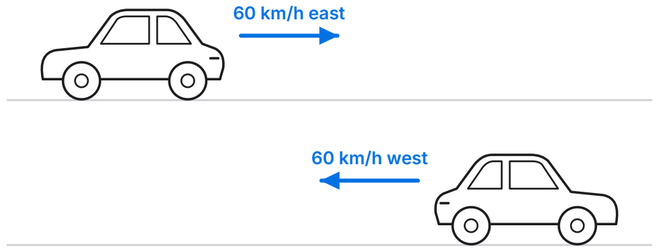
Speed with a direction.

DEFINITION

Velocity is the displacement per unit time. A vector.

$$\text{velocity} = \frac{\text{displacement}}{\text{time}}$$

Same SI unit as speed: **m/s**. But the answer must say which way.



Same speed. Different velocities, because the directions differ.

SAY

Point to the two cars: same speedometer reading, but they are going opposite ways. Their speeds are equal; their velocities are not.

ASK

If a car drives round a roundabout at a steady 30 km/h, is its velocity changing? (Yes: the direction keeps changing.)

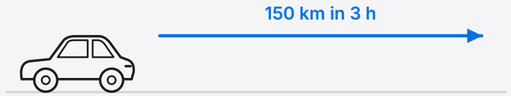
MISCONCEPTION TO ADDRESS

Students use speed and velocity as the same word. Whenever a direction matters, use velocity.

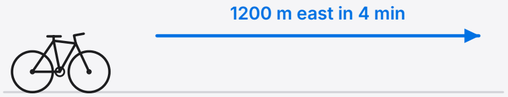
Minutes 23–30 · 2 min

WORKED EXAMPLES

Speed and velocity in numbers.



QUESTION	A car covers 150 km in 3 h. Find its average speed.
FORMULA	$v = \frac{s}{t}$
SOLVE	$\frac{150\ \text{km}}{3\ \text{h}} = 50\ \text{km/h}$
ANSWER	$50\ \text{km/h}$ **or about** $13.9\ \text{m/s}$



QUESTION	A cyclist rides 1200 m due east in 4 minutes. Find her velocity.
CONVERT	$4\ \text{min} = 4 \times 60 = 240\ \text{s}$
SOLVE	$\frac{1200\ \text{m}}{240\ \text{s}} = 5\ \text{m/s}$
ANSWER	$5\ \text{m/s}$ **due east**

SAY

Work the first one quickly. In the second, slow down at the conversion: minutes must become seconds before dividing, because the answer must be in m/s.

MISCONCEPTION TO ADDRESS

Forgetting to convert minutes to seconds. And forgetting the direction in a velocity answer.

05 Speed and velocity

Minutes 23–30

AVERAGE SPEED

The trap almost everyone falls into.

$$\text{average speed} = \frac{\text{total distance}}{\text{total time}}$$

QUESTION	Go 60 km at 60 km/h. Return the same 60 km at 30 km/h. Average speed?
TIMES	Going: 1 h. Returning: 2 h. Total: 3 h.
SOLVE	$\frac{120\ \text{km}}{3\ \text{h}}$
ANSWER	$40\ \text{km/h}$ **(not 45 km/h)**

ASK THE CLASS

Why is the answer not simply the average of 60 and 30?

Remember: never average the speeds. Always divide total distance by total time. The slow part takes longer, so it counts for more.

SAY

Let the class guess first. Most will say 45. Then work it. This surprise makes the rule stick.

ASK

Why is the answer not 45? (Because the car spends twice as long at 30 km/h as at 60 km/h, so the slower speed counts for more of the time.)

MISCONCEPTION TO ADDRESS

Averaging the speeds is the single most common error in average speed questions.

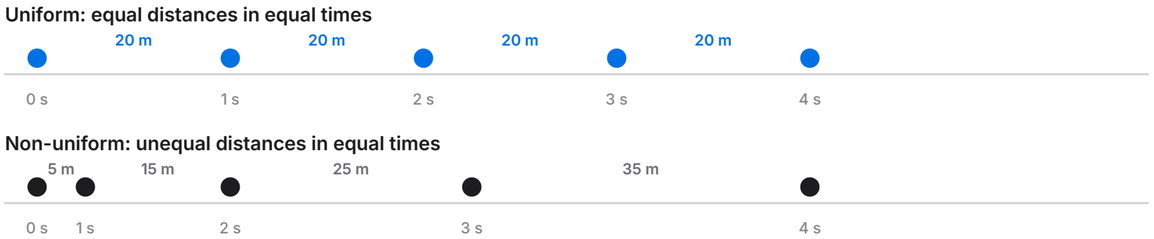
Minutes 30–35 · 2 min

06 Uniform and non-uniform motion

Minutes 30–35

06 · UNIFORM AND NON-UNIFORM MOTION

Is the speed steady, or changing?



UNIFORM MOTION

Equal distances in equal intervals of time. A car on cruise control.

NON-UNIFORM MOTION

Unequal distances in equal intervals. A bus leaving a stop.

SAY

Each dot is a photograph taken every second. Equal gaps means steady speed. Gaps that grow mean the object is speeding up. Ask which describes traffic in a city.

ASK

Is a ball rolling to a stop uniform or non-uniform? (Non-uniform: gaps get smaller.)

MISCONCEPTION TO ADDRESS

Students think 'uniform' means 'moving fast'. It means moving at a steady rate in a straight line.

Minutes 30–35 · 3 min

06 Uniform and non-uniform motion

Minutes 30–35

DISTANCE–TIME GRAPHS

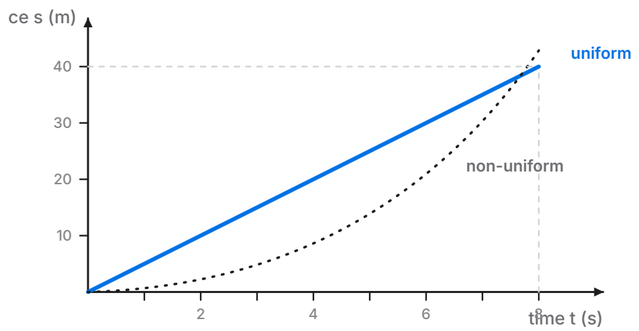
The slope tells you the speed.

A straight line means **uniform** speed. A curve means the speed is **changing**. The steeper the line, the faster the object.

READ The blue line reaches 40 m at 8 s.

SOLVE $v = \frac{40\ \text{m}}{8\ \text{s}}$

ANSWER $5\ \text{m/s}$



SAY

Show how to read two values off the graph. Then trace the curve with your finger: it gets steeper, so the object is speeding up.

ASK

A flat horizontal line on a distance–time graph means what? (The object is at rest: distance is not changing.)

MISCONCEPTION TO ADDRESS

Students read a distance–time graph as a picture of the path, like a hill. It is not a map; it shows how distance changes with time.

Minutes 35–40 · 2 min

07 Acceleration

Minutes 35–40

07 · ACCELERATION

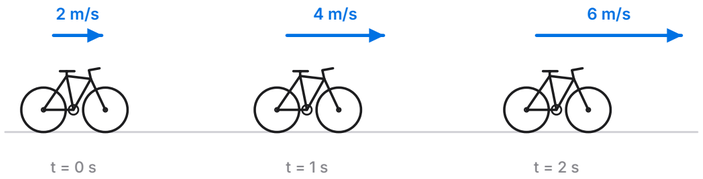
How quickly velocity changes.

DEFINITION

Acceleration is the change in velocity per unit time. A vector.

$$a = \frac{v - u}{t}$$

SYMBOL	MEANS	SI UNIT
a	acceleration	m/s²
u	initial velocity	m/s
v	final velocity	m/s
t	time taken	s



Velocity rises by 2 m/s every second: $a = 2\ \mathrm{m/s^2}$

SAY

Walk through the three bicycles: every second, the velocity goes up by the same 2 m/s. That steady change is acceleration. Read the unit aloud: 'metres per second, per second'.

MISCONCEPTION TO ADDRESS

Students confuse speed and acceleration: 'fast' is not 'accelerating'. A car cruising at 100 km/h in a straight line has zero acceleration.

Minutes 35–40 · 3 min

WORKED EXAMPLES

Speeding up and slowing down.

QUESTION	A car starts from rest and reaches 20 m/s in 5 s. Find its acceleration.
GIVEN	$u = 0$, $v = 20\ \text{m/s}$, $t = 5\ \text{s}$
FORMULA	$a = \frac{v-u}{t}$
SOLVE	$\frac{20-0}{5}$
ANSWER	$a = 4\ \text{m/s}^2$

QUESTION	A cyclist brakes from 6 m/s to rest in 3 s. Find the acceleration.
GIVEN	$u = 6\ \text{m/s}$, $v = 0$, $t = 3\ \text{s}$
FORMULA	$a = \frac{v-u}{t}$
SOLVE	$\frac{0-6}{3}$
ANSWER	$a = -2\ \text{m/s}^2$, slowing down

ASK THE CLASS

What does the minus sign in the second answer tell us?

SAY

Stress 'starts from rest' means u = 0: exam questions use this phrase constantly. In the second example, let the minus sign appear naturally and ask what it means.

ASK

What does the minus sign mean? (The acceleration is opposite to the motion, so the cyclist slows down. This is called retardation or deceleration.)

MISCONCEPTION TO ADDRESS

Students think negative acceleration always means slowing down. It means acceleration in the negative direction; it slows the object when it is opposite to the velocity, as here.

Minutes 40–48 · 2 min

08 · EQUATIONS OF MOTION

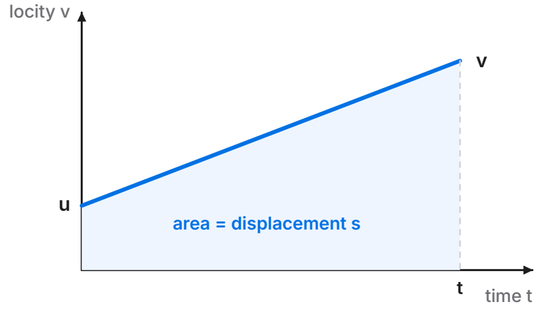
Three equations for uniform acceleration.

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

They work only when the acceleration is **uniform** (constant) and the motion is in a straight line.



On a velocity–time graph, the slope is a and the area under the line is s.

SAY

Show where the first equation comes from: rearrange a = (v − u)/t. Mention that the second comes from the shaded area on the graph. The textbook derives all three from this graph; today we focus on using them.

MISCONCEPTION TO ADDRESS

Students apply these equations when acceleration is not constant. Always check the question says 'uniform acceleration' or 'uniformly'.

Minutes 40–48 · 1.5 min

08 Equations of motion

Minutes 40–48

KNOW YOUR LETTERS

Five quantities. Each equation leaves one out.

SYMBOL	MEANS	SI UNIT
u	initial velocity	m/s
v	final velocity	m/s
a	acceleration	m/s²
t	time	s
s	displacement	m

EQUATION	LEAVES OUT
$v = u + at$	s
$s = ut + \frac{1}{2}at^2$	v
$v^2 = u^2 + 2as$	t

THE METHOD

1. Write what you know. 2. Write what you need. 3. Pick the equation that does not use the one quantity you neither know nor need.

SAY

This slide is the key skill. Teach the 'leave one out' trick: every problem gives three quantities and asks for a fourth; choose the equation without the fifth.

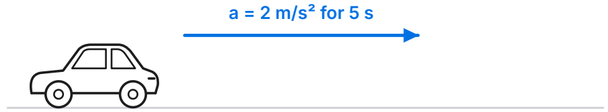
MISCONCEPTION TO ADDRESS

Students try all three equations at random. Insist on writing 'Given' and 'Find' first, every time.

Minutes 40–48 · 2 min

WORKED EXAMPLE 1

A car pulls away.



A car starts from rest and accelerates uniformly at $2\ \mathrm{m/s^2}$ for $5\ \mathrm{s}$. Find its final velocity and the distance covered.

GIVEN	$u = 0,\ a = 2\ \mathrm{m/s^2},\ t = 5\ \mathrm{s}$
VELOCITY	$v = u + at = 0 + 2 \times 5 = 10\ \mathrm{m/s}$
DISTANCE	$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 2 \times 5^2 = 25\ \mathrm{m}$
CHECK	$v^2 = 2as \Rightarrow 100 = 2 \times 2 \times 25$ ✓
ANSWER	$v = 10\ \mathrm{m/s},\ s = 25\ \mathrm{m}$

SAY

Model the layout students should copy: Given, formula, substitution, answer with unit. Show the check with the third equation: it builds confidence that the answers are right.

MISCONCEPTION TO ADDRESS

Squaring errors: ½ × 2 × 5² is ½ × 2 × 25, not ½ × 2 × 10.

Minutes 40–48 · 2.5 min

WORKED EXAMPLE 2

A train brakes.

A train moving at $20\ \mathrm{m/s}$ brakes with a uniform acceleration of $-0.5\ \mathrm{m/s^2}$. How far does it travel before stopping, and how long does it take?

ASK THE CLASS

Which equation finds the distance without knowing the time?

GIVEN	$u = 20\ \mathrm{m/s},\ v = 0,\ a = -0.5\ \mathrm{m/s^2}$
DISTANCE	$v^2 = u^2 + 2as \Rightarrow 0 = 400 + 2(-0.5)s$
	$s = \frac{400}{1} = 400\ \mathrm{m}$
TIME	$v = u + at \Rightarrow 0 = 20 - 0.5t$
	$t = \frac{20}{0.5} = 40\ \mathrm{s}$
ANSWER	$s = 400\ \mathrm{m},\ t = 40\ \mathrm{s}$

SAY

Ask the question on the slide before solving. The answer is the third equation, because time is unknown. Then find the time with the first.

ASK

Which equation? ($v^2 = u^2 + 2as$, because it does not contain t.)

MISCONCEPTION TO ADDRESS

Sign errors: braking means a is negative. Students who use +0.5 get a negative distance and get confused.

Minutes 48–57 · 1.5 min

09 Force and Newton's laws

Minutes 48–57

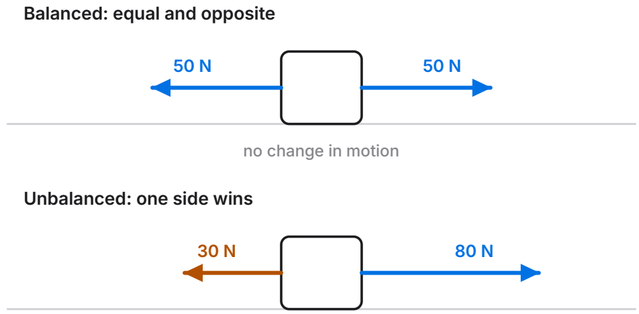
09 · FORCE

A push or a pull.

DEFINITION

A **force** is a push or pull that can change an object's **speed**, **direction** or **shape**. A vector, measured in **newtons (N)**.

Only an **unbalanced** (net) force changes motion.



SAY

Push a book across the desk. Then push it from both sides equally: nothing happens. That is balanced versus unbalanced. In the second picture the net force is 80 − 30 = 50 N to the right.

ASK

In the second picture, what is the net force? (50 N to the right.)

MISCONCEPTION TO ADDRESS

Students think a moving object must have a force pushing it forwards. Introduce the idea here; the first law settles it.

Minutes 48–57 · 2 min

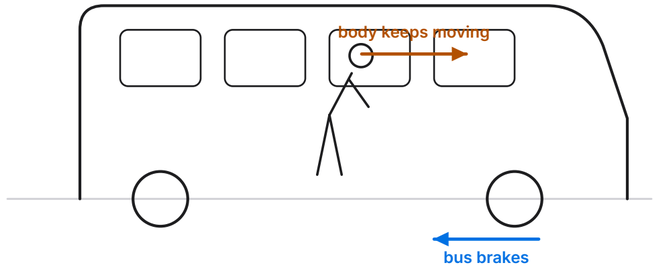
NEWTON'S FIRST LAW

Objects keep doing what they are doing.

FIRST LAW (LAW OF INERTIA)

An object stays at rest, or keeps moving in a straight line at a steady speed, unless an unbalanced force acts on it.

Inertia is this tendency to resist a change in motion. More mass means more inertia.



When the bus brakes, your body tends to keep moving forwards.

SAY

Everyone has lurched forward in a braking bus: the bus stopped, their body did not want to. If possible, do the coin-on-card demonstration: flick the card and the coin drops into the glass.

ASK

Why does a ball rolling on grass eventually stop, if no one is pushing back? (Friction is an unbalanced force acting against its motion.)

MISCONCEPTION TO ADDRESS

The big misconception: 'objects need a force to keep moving'. They need a force to change motion. Things stop because friction acts on them.

Minutes 48–57 · 1.5 min

Minutes 48–57

NEWTON'S SECOND LAW

Force, mass and acceleration.

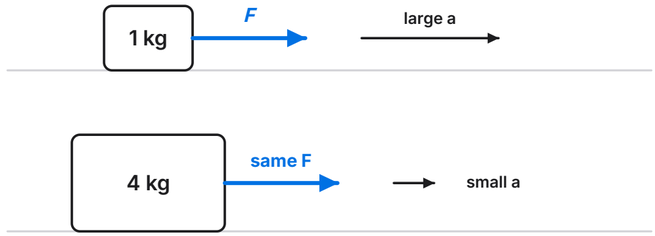
SECOND LAW

The acceleration of an object is directly proportional to the net force and inversely proportional to its mass.

$$F = ma$$

SYMBOL	MEANS	SI UNIT
F	net force	newton (N)
m	mass	kg
a	acceleration	m/s²

$1\ \mathrm{N} = 1\ \mathrm{kg\ m/s^2}$



SAY

Use the trolley picture: the same push on a light trolley and a heavy one. The light one speeds up faster. Then show F = ma as the short form of that sentence. If time allows, add that the textbook states the law using momentum, p = mv: force is the rate of change of momentum, which gives F = ma when mass stays constant.

ASK

Which is harder to stop, a cricket ball or a football moving at the same speed? Why? (The football usually has more mass, so more momentum and more inertia.)

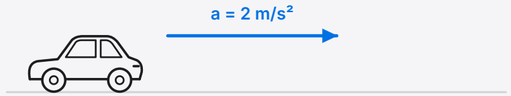
MISCONCEPTION TO ADDRESS

Students use mass in grams. Mass must be in kilograms for force to come out in newtons.

Minutes 48–57 · 1.5 min

WORKED EXAMPLES

Using F = ma.



QUESTION	What force gives a 1000 kg car an acceleration of 2 m/s²?
FORMULA	$F = ma$
SOLVE	1000×2
ANSWER	$F = 2000\ \text{N}$



QUESTION	A net force of 10 N acts on a 0.5 kg ball. Find its acceleration.
FORMULA	$a = \frac{F}{m}$
SOLVE	$\frac{10}{0.5}$
ANSWER	$a = 20\ \text{m/s}^2$

SAY

Do both quickly. In the second, show the rearrangement a = F/m as a separate line.

MISCONCEPTION TO ADDRESS

Dividing 10 by 0.5 gives 20, not 5. Students who rush often divide the wrong way.

Minutes 48–57 · 1.5 min

09 Force and Newton's laws

Minutes 48–57

NEWTON'S THIRD LAW

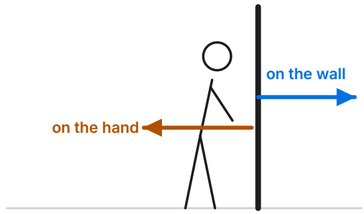
Forces come in pairs.

THIRD LAW

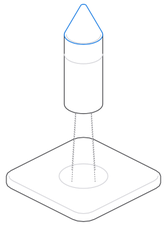
For every action there is an equal and opposite reaction.

The two forces act on **different objects**, so they never cancel each other out.

Walking: your foot pushes the ground back, the ground pushes you forward. Swimming, a gun's recoil and a rocket all work the same way.



Two objects, two forces



Gases pushed down, rocket pushed up

SAY

Ask a student to push the wall. Can they feel the wall pushing back? Then the rocket: it pushes hot gases down, the gases push the rocket up.

ASK

If action and reaction are equal and opposite, why do they not cancel? (They act on different objects: one on the wall, one on the hand.)

MISCONCEPTION TO ADDRESS

The biggest third-law misconception is that the pair cancels. Forces cancel only when they act on the same object.

Minutes 48–57 · 1 min

09 Force and Newton's laws

Minutes 48–57

YOUR TURN

Which law is it?

SITUATION A
A tablecloth is pulled quickly and the dishes stay on the table.

SITUATION B
Kicking a football harder makes it speed up more.

SITUATION C
A swimmer pushes water backwards and moves forwards.

SAY

Read each situation. Students answer 1, 2 or 3 with fingers. Ask one student to explain each answer in one sentence.

ASK

Answers: A: first law (inertia of the dishes). B: second law (bigger force, bigger acceleration). C: third law (action and reaction).

Minutes 57–60 · 1.5 min

PRACTICE

CBSE-style questions.

Try Q1 and Q2 now. The rest are for homework; full solutions follow.

Q1 CONCEPTUAL

Can an object have zero displacement but a non-zero distance? Give an example.

Q2 NUMERICAL

Convert 54 km/h into m/s, and 25 m/s into km/h.

Q3 NUMERICAL

A bus starts from rest and accelerates uniformly at 0.1 m/s² for 2 minutes. Find its final speed and the distance it travels.

Q4 NUMERICAL

A train moving at 90 km/h brakes with a uniform acceleration of −0.5 m/s². How far does it travel before it stops?

SAY

Give two minutes for Q1 and Q2 in class. Set Q3 to Q8 as homework. The answer key follows these two slides; share it after students have attempted the questions.

ASK

Q1 answer: yes, e.g. one lap of a track. Q2: 15 m/s and 90 km/h.

10 Practice and recap

Homework

PRACTICE

CBSE-style questions, continued.

Q5 NUMERICAL

An athlete runs round a circular track of diameter 200 m and completes one round in 40 s. Find the distance and the displacement after 2 min 20 s. (Take π = 22/7.)

Q6 NUMERICAL

What net force is needed to give a 1500 kg car an acceleration of 2 m/s²?

Q7 CONCEPTUAL

Passengers in a bus fall forward when it stops suddenly. Explain why, using Newton's first law.

Q8 CONCEPTUAL

Action and reaction forces are equal and opposite. Explain why they do not cancel each other.

SAY

Homework questions. Point out that Q5 is a classic NCERT-style problem: it combines distance, displacement and a circle.

10 Practice and recap

Answer key

ANSWER KEY

Detailed solutions, Q1 to Q4.

Q1 Yes. Distance is the length of the path, displacement is the straight line from start to finish. An athlete who runs one full lap of a track covers a distance of 400 m, but ends at the start, so the displacement is 0.

Q2 $54 \times \frac{5}{18} = 15\ \text{m/s}$. $25 \times \frac{18}{5} = 90\ \text{km/h}$.

Q3 $u = 0,\ a = 0.1\ \text{m/s}^2,\ t = 2\ \text{min} = 120\ \text{s}$. $v = u + at = 0.1 \times 120 = 12\ \text{m/s}$. $s = \frac{1}{2}at^2 = \frac{1}{2} \times 0.1 \times 120^2 = 720\ \text{m}$.

Q4 $u = 90 \times \frac{5}{18} = 25\ \text{m/s}, v = 0, a = -0.5\ \text{m/s}^2$. $v^2 = u^2 + 2as \Rightarrow 0 = 625 - s$, so $s = 625\ \text{m}$.

SAY

Share after the homework is attempted.

10 Practice and recap

Answer key

ANSWER KEY

Detailed solutions, Q5 to Q8.

Q5 $2\ \text{min}\ 20\ \text{s} = 140\ \text{s}$, so rounds $= \frac{140}{40} = 3.5$. Circumference $= \pi d = \frac{22}{7} \times 200 = \frac{4400}{7}\ \text{m}$. Distance $= 3.5 \times \frac{4400}{7} = 2200\ \text{m}$. After 3.5 rounds the athlete is on the opposite side, so displacement $= d = 200\ \text{m}$.

Q6 $F = ma = 1500 \times 2 = 3000\ \text{N}$.

Q7 The bus stops because the brakes act on it, but the passengers' bodies tend to stay in motion because of inertia (Newton's first law), so they move forward.

Q8 Action and reaction act on two different objects. Forces can cancel only when they act on the same object, so the pair does not cancel.

SAY

For Q5, draw the circle and mark where 3.5 rounds ends: directly opposite the start, one diameter away.

Minutes 57–60 · 1 min

10 Practice and recap

Minutes 57–60

RECAP

Nine ideas to take home.

01
Physical quantity
a number and a unit

02
Scalar / vector
size only / size and direction

03
Distance / displacement
path length / straight line, with direction

04
Speed / velocity
distance per time / displacement per time

05
Uniform motion
equal distances in equal times

06
Acceleration
change in velocity per unit time

07
First law
inertia: no change without a net force

08
Second law
F = ma

09
Third law
equal and opposite pairs, on different objects

SAY

Ask a different student to say each card in their own words. If anyone hesitates, that is tomorrow's starter.

Minutes 57–60 · 0.5 min

Minutes 57–60

FORMULA SHEET

Everything you need for today's questions.

$v = \frac{s}{t}$ speed	$\text{avg speed} = \frac{\text{total distance}}{\text{total time}}$ average speed	$a = \frac{v-u}{t}$ acceleration	$v = u + at$ first equation
$s = ut + \frac{1}{2}at^2$ second equation	$v^2 = u^2 + 2as$ third equation	$F = ma$ Newton's second law	$1\ \text{km/h} = \frac{5}{18}\ \text{m/s}$ conversion

Equations of motion apply only to uniform acceleration in a straight line. Always use SI units: m, kg, s.

SAY

Students can photograph this slide. Remind them which formulas need uniform acceleration.

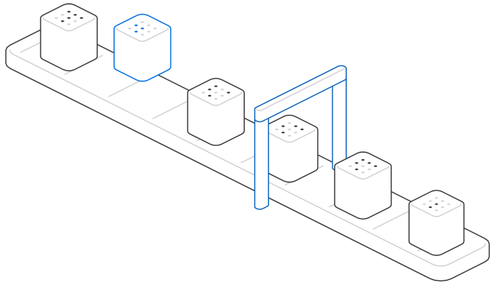
10 Practice and recap

Minute 60

NEXT LESSON

Motion in graphs, and circular motion.

Homework: practice questions 3 to 8. Bring your answers; we will check them together using the answer key.



SAY

Close by linking forward: velocity–time graphs and why moving in a circle at steady speed still counts as acceleration.